PROBLEM 14

Conservative Force and Potential V

Original statement · Problems.pdf, p. 9

Phase Analysis

A particle of mass m is moving along the Ox axis under the force $\mathbf{F}(x) = mKxe^{x/a}\mathbf{i}$, where K and a are known positive constants. The initial position of the particle is $x = a$ and the initial velocity is $\mathbf{v} = -v_0\mathbf{i}$. Describe the possible types of motion of the particle as a function of v_0.

Worked Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical System:** A particle of mass m moves along the Ox axis under a position-dependent non-linear force:

$$\mathbf{F}(x) = mKxe^{x/a}\mathbf{i}$$

- **Initial State:** At $t = 0$, the particle starts from $x(0) = a$ with initial velocity directed towards the negative axis:

$$\mathbf{v}(0) = -v_0\mathbf{i} \implies \dot{x}(0) = -v_0 < 0$$

- **Parameters:** $m > 0$ [kg], $K > 0$ [s^{-2}], $a > 0$ [m], $v_0 > 0$ [m/s].

PHASE 2

Reference Frames, Vector Bases & Coordinate Geometry

- **Inertial Frame:** Fixed Cartesian frame $\mathcal{F}_0 = \{O; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$.
- **Potential Energy Calculation via Integration by Parts:**

Pedagogical Justification: The force is 1D and depends only on x, so $\mathbf{F} = -\frac{dV}{dx}\mathbf{i}$. To find $V(x)$, we perform integration by parts on $\int xe^{x/a}\,dx$.

$$V(x) = -\int F(x)\,dx = -mK\int xe^{x/a}\,dx$$

Let:

$$u = x \implies du = dx$$

$$dv = e^{x/a}\,dx \implies v = ae^{x/a}$$

Applying the integration by parts formula $\int u\,dv = uv - \int v\,du$:

$$\int xe^{x/a}\,dx = axe^{x/a} - \int ae^{x/a}\,dx = axe^{x/a} - a^2e^{x/a} = a^2e^{x/a}\left(\frac{x}{a} - 1\right)$$

Therefore:

$$V(x) = -mKa^2e^{x/a}\left(\frac{x}{a} - 1\right) + C$$

We choose the arbitrary integration constant $C = 0$ so that at the initial position $x = a$:

$$V(a) = -mKa^2e^1(1-1) = 0$$

This normalizes the potential so that $V(a) = 0$.

PHASE 3

Step-by-Step Mathematical Deduction

1. Topography of the Potential $V(x)$

- First derivative:

$$\frac{dV}{dx} = -F(x) = -mKxe^{x/a}$$

Setting $\frac{dV}{dx} = 0$ yields a single critical point at $x = 0$.

- Second derivative:

$$\frac{d^2V}{dx^2} = -mK\frac{d}{dx}(xe^{x/a}) = -mK\left(e^{x/a} + \frac{x}{a}e^{x/a}\right) = -mKe^{x/a}\left(1 + \frac{x}{a}\right)$$

At $x = 0$:

$$\frac{d^2V}{dx^2}(0) = -mK(1) = -mK < 0 \implies \textbf{Local Maximum}$$

The value of the potential barrier at the origin is:

$$V_{max} = V(0) = -mKa^2e^0(0-1) = +mKa^2$$

- Asymptotic behaviors:
- As $x \to -\infty$: $e^{x/a} \to 0$, so $V(-\infty) = 0$.

- As $x \to +\infty$: $-mKa^2e^{x/a}(x/a - 1) \to -\infty$.

2. Total Mechanical Energy E

Since $V(a) = 0$, the total energy of the particle is entirely kinetic at $t = 0$:

$$E = T(0) + V(a) = \frac{1}{2}mv_0^2 + 0 = \frac{1}{2}mv_0^2$$

3. Determination of the Critical Velocity $v_{0,crit}$

> **Pedagogical Justification:** The particle is launched leftwards from $x = a > 0$ towards the barrier at $x = 0$. To reach the summit of the barrier at $x = 0$, its total mechanical energy must at least equal the barrier height $V(0)$:

$$E = V(0) \implies \frac{1}{2}mv_{0,crit}^2 = mKa^2$$

Multiplying by $2/m$:

$$v_{0,crit}^2 = 2Ka^2 \implies v_{0,crit} = a\sqrt{2K}$$

4. Classification of Motion Regimes

1. **Regime 1:** $v_0 < a\sqrt{2K}$ **(**$E < V(0)$**)**

- The initial kinetic energy is insufficient to overcome the potential barrier at $x = 0$.
- As the particle moves towards the origin, it decelerates because $F(x) = mKxe^{x/a} > 0$ for all $x > 0$.
- The particle comes to a momentary stop at a turning point $x_{turn} \in (0, a)$, where $V(x_{turn}) = E$.
- Since $F(x_{turn}) > 0$, it is accelerated back towards the positive direction, passes $x = a$, and accelerates indefinitely towards $+\infty$.
- **Conclusion:** The particle decelerates before arriving at $x = 0$, stops, and moves towards $+\infty$.

2. **Regime 2:** $v_0 = a\sqrt{2K}$ **(**$E = V(0)$**)**

- The particle has precisely the energy needed to reach the top of the barrier at $x = 0$.
- Near $x = 0$, Taylor expanding the potential:

$$V(x) \approx V(0) + V'(0)x + \frac{1}{2}V''(0)x^2 = mKa^2 - \frac{1}{2}mKx^2$$

The velocity as a function of position is:

$$\dot{x} = -\sqrt{\frac{2}{m}[E - V(x)]} \approx -\sqrt{\frac{2}{m}\left[mKa^2 - \left(mKa^2 - \frac{1}{2}mKx^2\right)\right]} = -\sqrt{K}\,x$$

Separating variables to find the transit time to $x = 0$:

$$\int_a^0 \frac{dx}{x} = -\sqrt{K}\int_0^t dt' \implies [\ln x]_a^0 = -\infty$$

- **Conclusion:** The transit time diverges logarithmically. The particle takes **infinite time** to reach $x = 0$ (it asymptotically approaches $x = 0$ without ever reaching it).

3. **Regime 3:** $v_0 > a\sqrt{2K}$ **(**$E > V(0)$**)**

- The initial energy exceeds the barrier height $V(0)$.
- The particle reaches $x = 0$ with residual speed:

$$v(0) = \sqrt{\frac{2}{m}[E - V(0)]} = \sqrt{v_0^2 - 2Ka^2} > 0$$

- Upon crossing into $x < 0$, the force becomes negative ($F(x) = mKxe^{x/a} < 0$ for $x < 0$), accelerating the particle further to the left.
- **Conclusion:** The particle crosses the origin and moves towards $-\infty$.

PHASE 4

Physical Interpretation & Dimensional Verification

- Compare with the official solution key (Problems.pdf, Solution keys section):
- If $v_0 > a\sqrt{2K}$, particle moves towards $-\infty$.
- If $v_0 < a\sqrt{2K}$, particle moves towards $+\infty$ after decelerating before arriving to $x = 0$.
- If $v_0 = a\sqrt{2K}$, particle takes infinite time to reach $x = 0$.
- Dimensional verification: $[v_{0,crit}] = [a]\sqrt{[K]} = \mathrm{m} \cdot \sqrt{\mathrm{s}^{-2}} = \mathrm{m/s}$. Correct.
- **Dimensional check of the barrier:** $[mKa^2] = \mathrm{kg} \cdot \mathrm{s}^{-2} \cdot \mathrm{m}^2 = \mathrm{J}$, the same as $\frac{1}{2}mv_0^2$, so $v_0^2/(2Ka^2)$ is dimensionless and decides the regime.
- **Limits:** for $K \to 0$ the force vanishes, $v_{0,crit} \to 0$ and every $v_0 > 0$ crosses the origin (free particle); for $v_0 \gg a\sqrt{2K}$ the barrier is negligible, $v(0) \to v_0$.
- **Numerical check (RK4, step 10^{-4}, $m = K = a = 1$, $v_{0,crit} = 1.41421$):** for $v_0 = 1.2$ the particle turns at $x = 0.60801$ (root of $-e^x(x-1) = 0.72$ is 0.608007) and returns to $+\infty$; for $v_0 = 1.6$ it crosses the origin and reaches $x = -6$ with $\dot{x} = -1.5891$, equal to $-\sqrt{2(E - V(-6))} = -1.5891$ from energy conservation.

Source: Problems.pdf